calm_weighted

equity strategy | family: equity-volatility | status: validated | derived from `balanced_leverage`

What this strategy does

Balanced Leverage, with each bet divided by how jumpy the stock is

Why it should work. PEAD is a stronger, cleaner signal in low-vol stocks; size positions by inverse risk.

Design notes (from the strategy's source file).

Calm Weighted -- Balanced Leverage, with bigger bets on calmer stocks (sized by inverse idiosyncratic volatility).

Within every (quarter, side, sector, size) cell, each event's weight is scaled by 1/ivol, so calm stocks get more capital than wild ones. Rationale: the earnings drift is worth more per unit of risk in low-vol names (the SUE x vol interaction is t = -2.4 to -2.7 in Fama-MacBeth), and the engine's cost/liquidity model charges the same bps regardless of vol. Raises Sharpe 1.30 -> 1.47 with no fitted parameter, in all three sub-eras.

`CalmWeightedMatched` re-sizes it to Balanced Leverage's exact volatility to answer "at the SAME risk, how much more return": 7.31% vs 6.49%.

parameter	value
base_unit_fraction	0.003
max_gross_leverage	2.5
liquidity_cap_frac	0.05
return_basis	mktadj
cap_mode	priority
exposure_cap	open_book
beta_overlay	0.0

Performance

metric	calm weighted	market (CRSP VW)
Annualized return	5.64%	9.94%
Annualized volatility	4.51%	19.35%
Sharpe (NAV-pct)	1.24	0.59
Sortino	1.79	
Max drawdown	-17.5%	-55.5%
Calmar	0.32	
Final NAV ($10M start)	$29,109,274	
Worst day	-3.69%	
% positive days	50.6%	
Sharpe 1996-2001	2.03	
Sharpe 2002-2007	1.54	

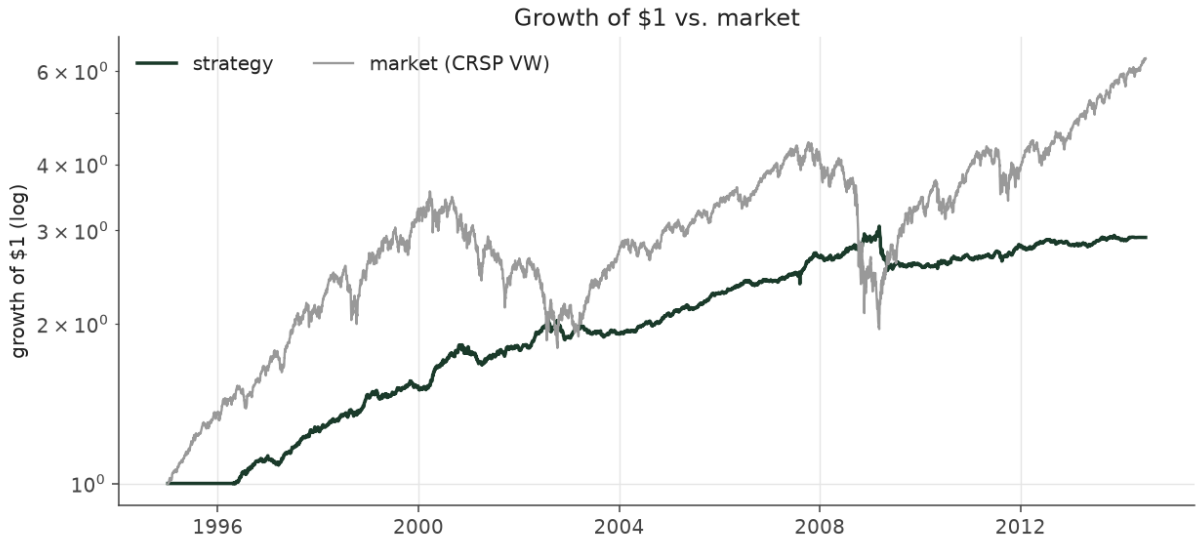
Sharpe 2008-2013	0.29	

Correlation of daily returns with the market: **-0.05**.

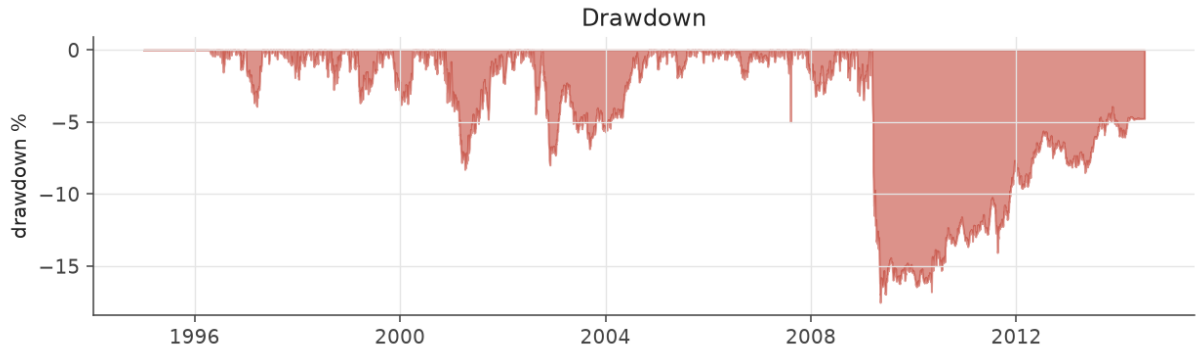
• best Sharpe among equity: `calm_rank_tilt` (1.81); this one: 1.24

• best return among equity: `market_boost_max` (0.15); this one: 0.0564

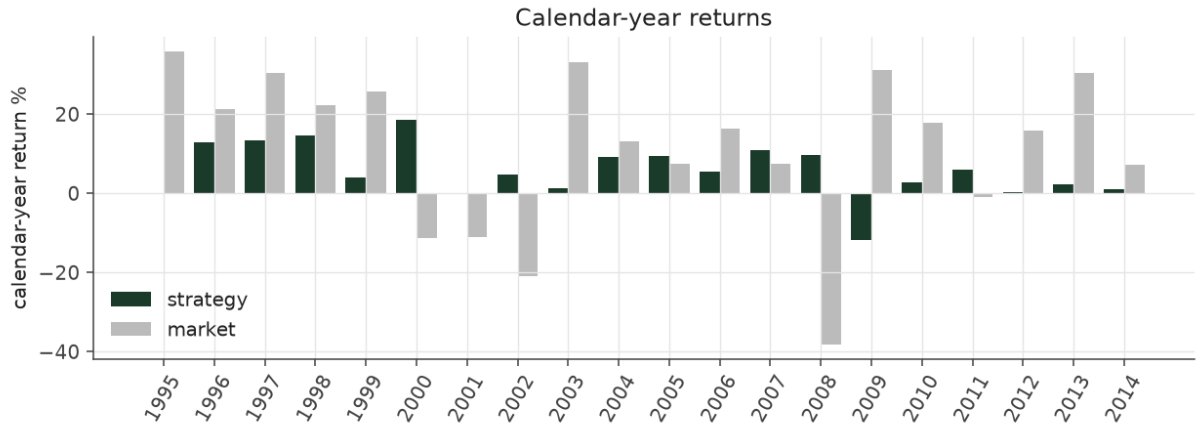
• best Calmar among equity: `calm_rank_tilt` (0.614); this one: 0.322



NAV (log scale)



Drawdown from running peak

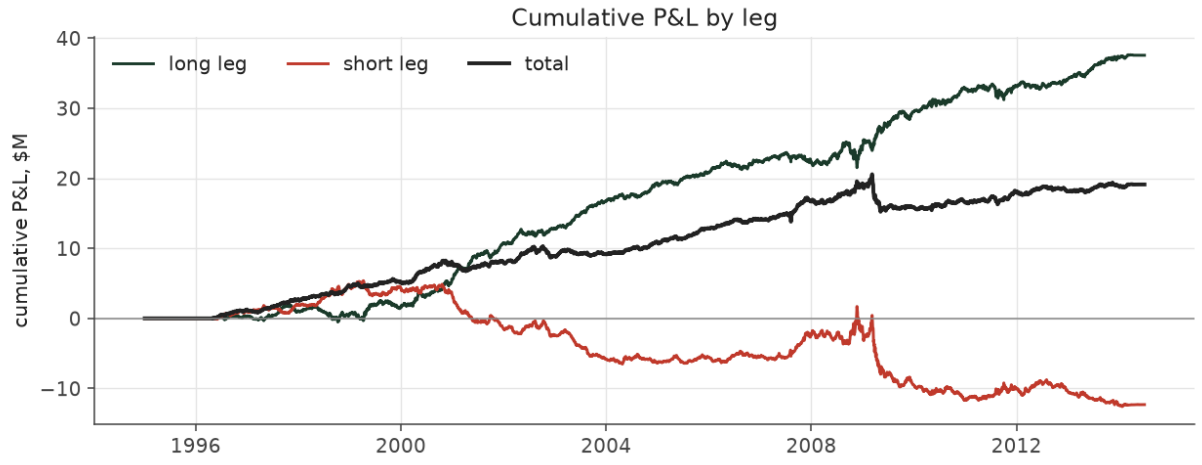


Calendar-year returns



Is the edge stable, or concentrated in a few years?

Where the return comes from



Long leg vs. short leg vs. total

	long leg	short leg
P&L ($M, before costs)	37.55	-12.31
P&L after costs ($M)	34.57	-15.46
of which common (average-event) component ($M)	23.96	-24.53

of which excess / PEAD component ($M)	13.59	12.22
excess minus costs ($M)	10.61	9.07
gross return per $ traded (bp)	252.23	-81.07
positions in the money (hit rate)	0.56	0.48
stand-alone Sharpe (leg only)	1.30	-0.32

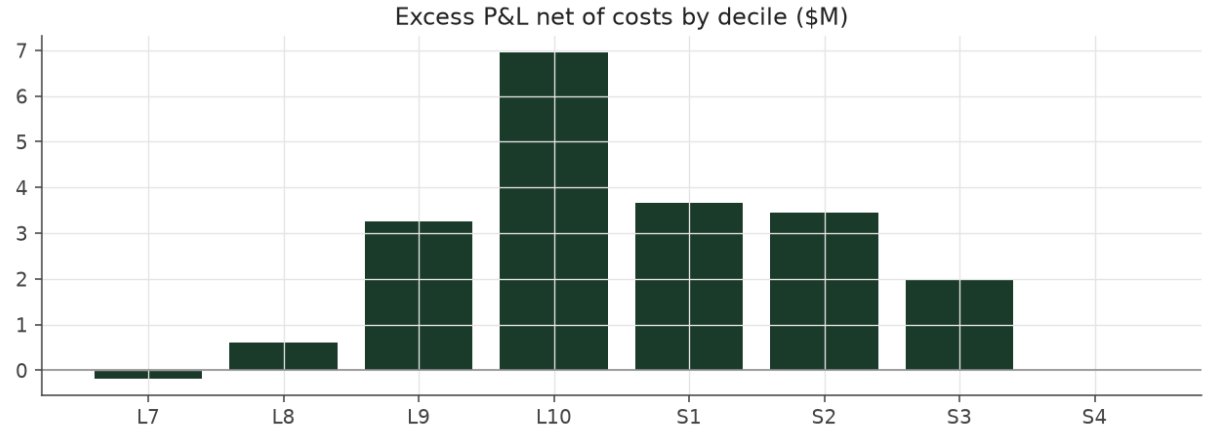
Daily P&L correlation between the two legs: **-0.79**.

• **The short leg loses money in absolute terms** (-12.3M before costs) while the long leg earns +37.5M. That is mostly a benchmark effect, not a signal failure: the average earnings-event stock beat the cap-weighted market during its holding window, which flatters the long leg and hurts the short leg by 24.0M / 24.5M. Stripping that common component, the surprise signal itself earned +13.6M on the long side and +12.2M on the short side.

• The two legs' daily P&L are -0.79 correlated: the short leg is the hedge that makes the long leg's variance disappear (stand-alone leg Sharpe 1.30 long / -0.32 short vs 1.24 combined).

By decile and side (signal-driven part only)

leg	decile	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
long	7	5,819	97.95	1.49	0.01	0.20	-0.19	-1.0%	-19.30
long	8	15,492	326.71	6.47	1.26	0.66	0.60	3.0%	18.36
long	9	17,756	457.99	9.69	4.16	0.91	3.25	16.5%	70.91
long	10	19,540	605.95	16.92	8.17	1.21	6.95	35.3%	114.73
short	1	17,672	600.75	-6.48	4.95	1.29	3.66	18.6%	60.97
short	2	18,038	522.42	-5.09	4.51	1.07	3.44	17.5%	65.93
short	3	14,145	352.94	-3.29	2.69	0.71	1.98	10.1%	56.08
short	4	2,187	42.32	-0.60	0.07	0.08	-0.01	-0.1%	-3.10



D10 and the short deciles carry the alpha; the middle of each leg earns little

By size quintile (1 = smallest)

size quintile	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
1	13,151	273.10	0.24	1.95	1.06	0.89	4.5%	32.69
2	21,121	562.95	6.30	8.24	1.60	6.64	33.7%	118.00

3	23,790	687.37	5.68	7.14	1.47	5.67	28.8%	82.55
4	25,693	736.76	4.21	5.16	1.18	3.98	20.2%	54.08
5	26,894	746.82	2.68	3.31	0.83	2.49	12.6%	33.33

By liquidity quintile (1 = least liquid)

liquidity quintile	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
1	12,462	206.58	2.24	3.68	1.03	2.64	13.4%	127.96
2	21,542	626.74	4.13	6.91	1.88	5.03	25.5%	80.21
3	24,096	750.06	6.10	7.81	1.50	6.31	32.1%	84.11
4	25,666	722.72	4.06	4.59	1.01	3.57	18.2%	49.45
5	26,879	700.65	2.52	2.77	0.70	2.07	10.5%	29.50

By holding-window length

hold bucket	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
<=40d	39,177	1,036.37	0.99	8.91	1.99	6.92	35.2%	66.78
41-60d	26,318	759.45	3.00	6.24	1.68	4.56	23.2%	60.04
61-90d	36,129	977.98	9.56	5.93	1.81	4.12	20.9%	42.12
>90d	9,025	233.22	5.55	4.73	0.65	4.08	20.7%	175.07

By era

era	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
1996-01	32,145	639.04	8.17	9.64	1.27	8.37	42.5%	130.97
2002-07	36,722	1,027.78	8.29	10.20	2.05	8.15	41.4%	79.27
2008-13	41,782	1,340.18	2.65	5.97	2.80	3.17	16.1%	23.63

By FF12 sector

ff12 sector	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
BusEq	19,826	287.79	1.29	1.86	0.62	1.25	6.3%	43.24
Chems	2,543	203.74	0.84	1.21	0.36	0.85	4.3%	41.63
Durbl	2,811	220.57	1.64	2.18	0.41	1.77	9.0%	80.14
Enrgy	6,234	247.93	2.85	3.35	0.47	2.88	14.6%	115.99
Hlth	10,915	287.33	1.18	1.80	0.61	1.19	6.1%	41.45
Manuf	11,462	268.82	1.44	2.10	0.57	1.53	7.8%	57.08
Money	15,495	249.53	2.01	2.66	0.58	2.08	10.6%	83.40
NoDur	4,886	247.29	1.11	1.54	0.48	1.06	5.4%	43.08
Other	16,535	274.79	2.20	2.90	0.62	2.27	11.5%	82.73
Shops	11,323	280.69	2.57	3.16	0.59	2.57	13.1%	91.61

Telcm	3,941	245.21	1.83	2.33	0.46	1.88	9.5%	76.50
Utils	4,678	193.31	0.14	0.71	0.36	0.35	1.8%	18.34

By idiosyncratic-volatility tercile

vol tercile	positions	notional M	net pnl M	excess M	cost M	alpha net M	alpha share	alpha bp per $
high vol	36,115	727.84	1.59	3.70	1.35	2.35	12.0%	32.36
low vol	37,603	1,259.70	9.68	12.76	2.74	10.01	50.9%	79.50
mid vol	35,206	972.34	7.05	8.47	1.93	6.54	33.2%	67.24

Short-borrow sensitivity (not modeled in the backtest)

borrow rate	fee M	ann. return	drag
0.0%	0.0	5.64%	-0.00%
0.5%	1.5	5.35%	0.29%
1.0%	3.1	5.04%	0.60%
2.0%	6.1	4.37%	1.27%
5.0%	15.3	1.69%	3.95%

The engine charges the same transaction cost on shorts as on longs and no stock-borrow fee. The short leg is concentrated in small, illiquid names, where borrow is scarce and expensive -- read the table as a floor on the cost of running the short leg.